EXTENSION BY ZERO COMMUTES WITH A PROPER MORPHISM OVER AN OPEN SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G4W (sources/stacks-duality/pinned-original-source.tex line 8472).

Allowed standard prerequisites (editorial).

- Extension by zero and its description through Deligne systems. Same chapter, Remark labelled remark-extension-by-zero, invoked in the statement, and Lemma labelled lemma-extension-by-zero, cited at proof lines 8494-8500. Fixes the functors Rj_! and the uniqueness used to reduce the claim.
- The 2-out-of-3 property for Deligne systems and the characterisation of the extension property. Same chapter, Lemmas labelled lemma-extension-by-zero-triangle, lemma-deligne-system-2-out-of-3 and lemma-characterize-extension-by-zero, cited at proof lines 8506-8510 and 8516-8520. Reduce along distinguished triangles and localise the final statement on Y.
- The algebraic characterisation of the property. Same chapter, Lemma labelled lemma-characterize-extension-by-zero-algebra, cited at proof lines 8520-8524. Turns the geometric statement into one about the inverse system of cohomology modules.
- Powers of an ideal and cohomology of a proper morphism. Cohomology of Schemes, Lemma labelled coherent-lemma-cohomology-powers-ideal-application, cited at proof lines 8528-8532. Supplies the finite A-module M and the pro-isomorphism.

Source reference. Original reference 0G4N: sources/stacks-duality/pinned-original-source.tex, source lines 8203–8203. Same-chapter pointer into the author environment remark at source line 8203; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0G4L: sources/stacks-duality/pinned-original-source.tex, source lines 8103–8103. Same-chapter pointer into the author environment lemma at source line 8103; invoked at proof line 8496 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4M: sources/stacks-duality/pinned-original-source.tex, source lines 8145–8145. Same-chapter pointer into the author environment lemma at source line 8145; invoked at proof line 8505 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4Q: sources/stacks-duality/pinned-original-source.tex, source lines 8236–8236. Same-chapter pointer into the author environment lemma at source line 8236; invoked at proof line 8506 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 08J5: sources/stacks-chow/tags.txt, source lines 10838–10838. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8514 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4T: sources/stacks-duality/pinned-original-source.tex, source lines 8358–8358. Same-chapter pointer into the author environment lemma at source line 8358; invoked at proof line 8524 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4S: sources/stacks-duality/pinned-original-source.tex, source lines 8320–8320. Same-chapter pointer into the author environment lemma at source line 8320; invoked at proof line 8527 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02OA: sources/stacks-chow/tags.txt, source lines 3509–3509. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8532 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of Noetherian schemes. Let $V \subset Y$ be an open subscheme and set $U = f^{-1}(V)$. Picture*

$$\begin{array}{ccc} U & \xrightarrow{j} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ V & \xrightarrow{j'} & Y \end{array}$$

Then we have a canonical isomorphism $Rj'_! \circ Rg_* \to Rf_* \circ Rj_!$ of functors $D^b_{Coh}(\mathcal{O}_U) \to$ Pro-$D^b_{Coh}(\mathcal{O}_Y)$ where $Rj_!$ and $Rj'_!$ are as in Remark 0G4N.

First proof. Let K be an object of $D^b_{Coh}(\mathcal{O}_U)$. Let (K_n) be a Deligne system for $U \to X$ whose restriction to U is constant with value K. Of course this means that (K_n) represents $Rj_!K$ in Pro-$D^b_{Coh}(\mathcal{O}_X)$. Observe that both $Rj'_!Rg_*K$ and $Rf_*Rj_!K$ restrict to the constant pro-object with value Rg_*K on V. This is immediate for the first one and for the second one it follows from the fact

that $(Rf_*K_n)|_V = Rg_*(K_n|_U) = Rg_*K$. By the uniqueness of Deligne systems in Lemma 0G4L it suffices to show that (Rf_*K_n) is pro-isomorphic to a Deligne system. The lemma referenced will also show that the isomorphism we obtain is functorial.

Proof that (Rf_*K_n) is pro-isomorphic to a Deligne system. First, we observe that the question is independent of the choice of the Deligne system (K_n) corresponding to K (by the aforementioned uniqueness). By Lemmas 0G4M and 0G4Q if we have a distinguished triangle

$$K \to L \to M \to K[1]$$

in $D^b_{Coh}(\mathcal{O}_U)$ and the result holds for K and M, then the result holds for L. Using the distinguished triangles of canonical truncations (Derived Categories, Remark 08J5) we reduce to the problem studied in the next paragraph.

Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Let $\mathcal{J} \subset \mathcal{O}_Y$ be a quasi-coherent sheaf of ideals cutting out $Y \setminus V$. Denote $\mathcal{J}^n\mathcal{F}$ the image of $f^*\mathcal{J}^n \otimes \mathcal{F} \to \mathcal{F}$. We have to show that $(Rf_*(\mathcal{J}^n\mathcal{F}))$ is a Deligne system. By Lemma 0G4T the question is local on Y. Thus we may assume $Y = \mathrm{Spec}(A)$ is affine and $\mathcal{J}$ corresponds to an ideal $I \subset A$. By Lemma 0G4S it suffices to show that the inverse system of cohomology modules $(H^p(X, I^n\mathcal{F}))$ is pro-isomorphic to the inverse system (I^nM) for some finite A-module M. This is shown in Cohomology of Schemes, Lemma 02OA. □